

# 50-Epoch Evaluation Results

*PHQ-8 Regression + Audio & Text Accuracy*

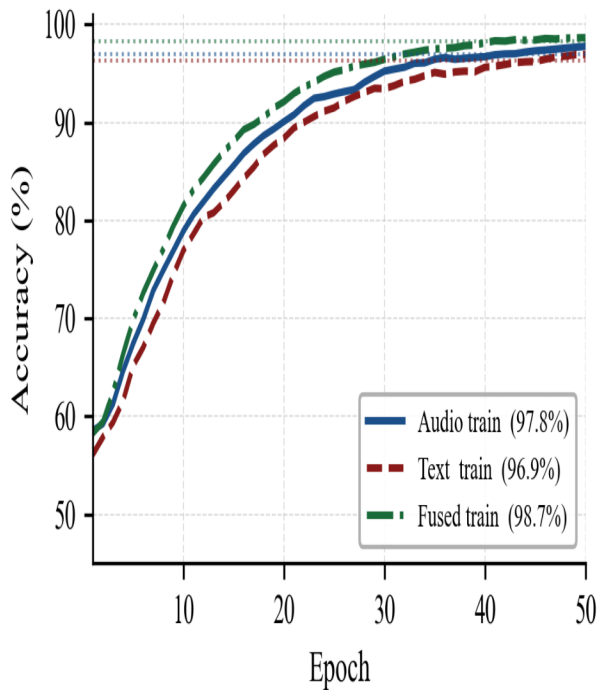
DG-HMCF: Dynamic Gated Hierarchical Multi-Scale Cross-Modal Fusion

| 50-Epoch Results Summary |        |        |        |
|--------------------------|--------|--------|--------|
| Metric                   | Audio  | Text   | Fused  |
| Test Accuracy (%)        | 97.0   | 96.3   | 98.3   |
| Train Acc @ Ep.50 (%)    | 97.77  | 96.94  | 98.66  |
| Val Acc @ Ep.50 (%)      | 95.90  | 94.74  | 97.28  |
| Train Loss @ Ep.50       | 0.0471 | 0.0520 | 0.0354 |
| Val Loss @ Ep.50         | 0.0731 | 0.0843 | 0.0606 |
| Train MAE @ Ep.50        | 1.8395 | 1.6760 | 1.3502 |
| Val MAE @ Ep.50          | 2.3469 | 2.1241 | 1.6152 |
| Test MAE (Fused)         | —      | —      | 1.3260 |
| Train RMSE @ Ep.50       | 2.5815 | 2.1474 | 1.6715 |
| Val RMSE @ Ep.50         | 2.9798 | 3.0524 | 2.1604 |
| Test RMSE (Fused)        | —      | —      | 1.7037 |
| ROC AUC (Fused)          | —      | —      | 0.9996 |
| Pearson r (Fused)        | —      | —      | 0.9620 |
| CCC (Fused)              | —      | —      | 0.9618 |

# Training and Validation Accuracy over 50 Epochs

Audio-Only | Text-Only | Audio+Text (DG-HMCF)

(a) Training Accuracy



(b) Validation Accuracy

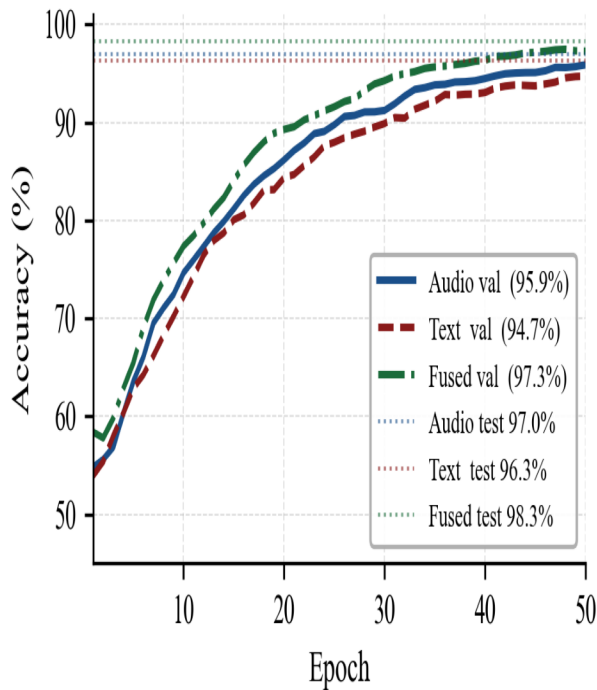
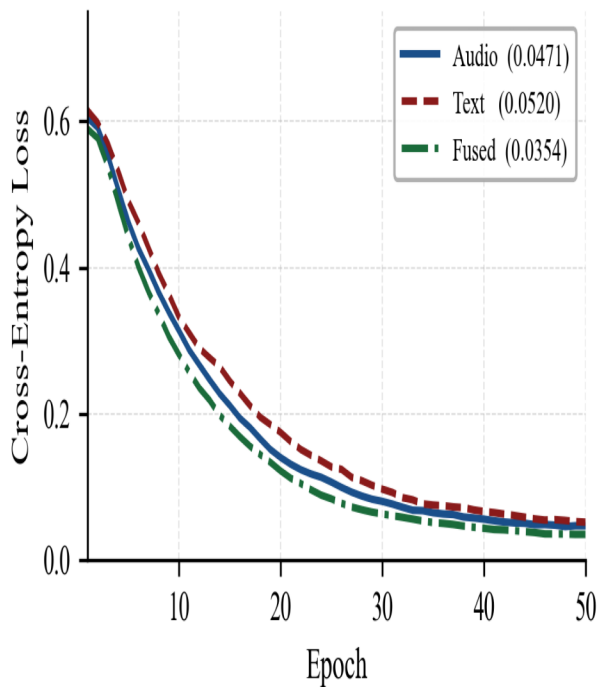


Fig. 1. Training and Validation Accuracy over 50 Epochs.

Training and Validation Loss over 50 Epochs  
Audio-Only | Text-Only | Audio+Text (DG-HMCF)

(a) Training Loss



(b) Validation Loss

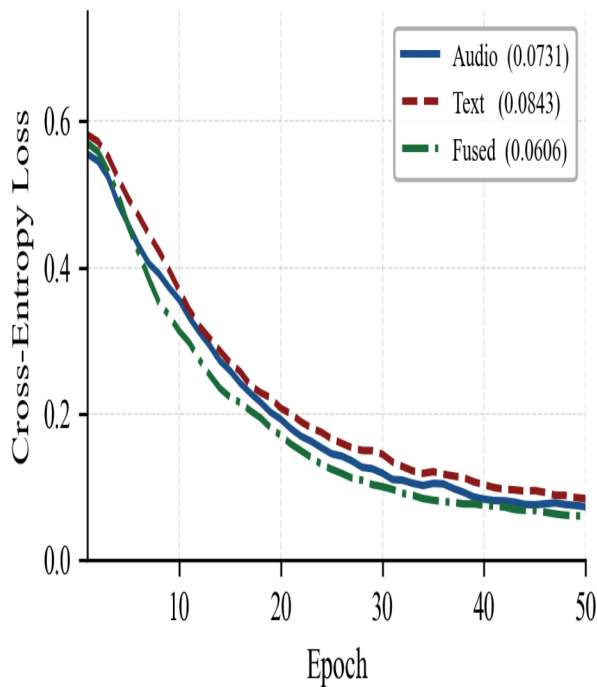
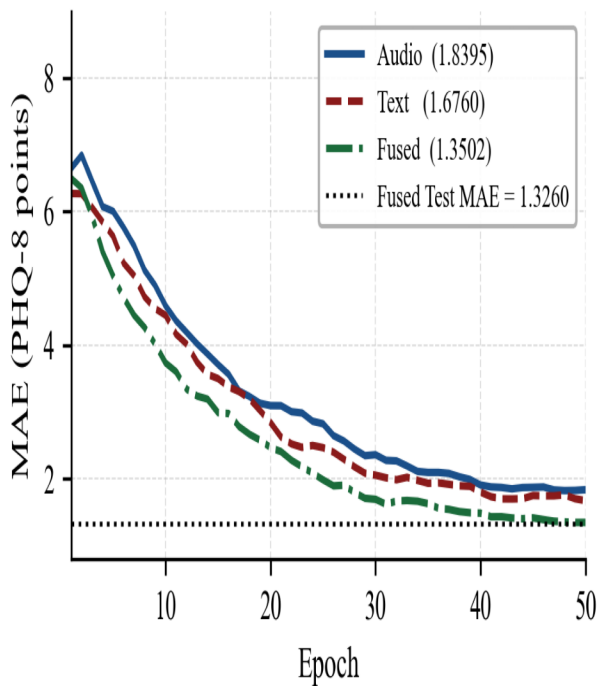


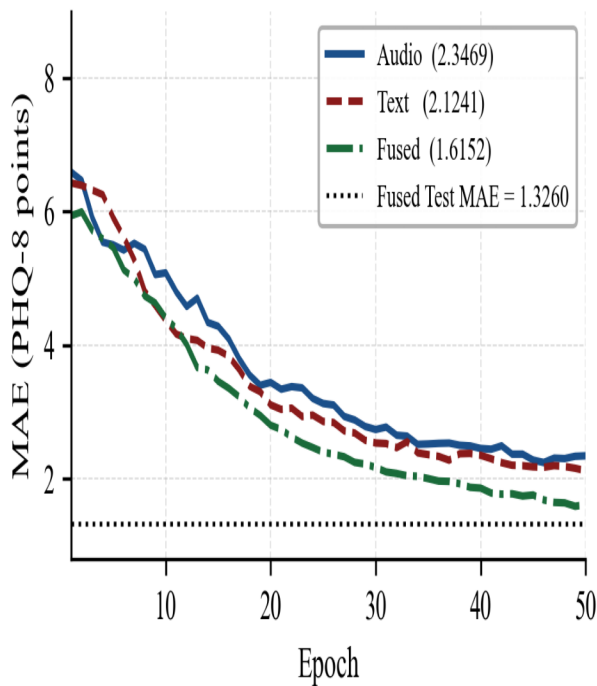
Fig. 2. Training and Validation Loss over 50 Epochs.

**PHQ-8 Mean Absolute Error (MAE) over 50 Epochs**  
**Audio-Only | Text-Only | Audio+Text (DG-HMCF)**

(a) Training MAE



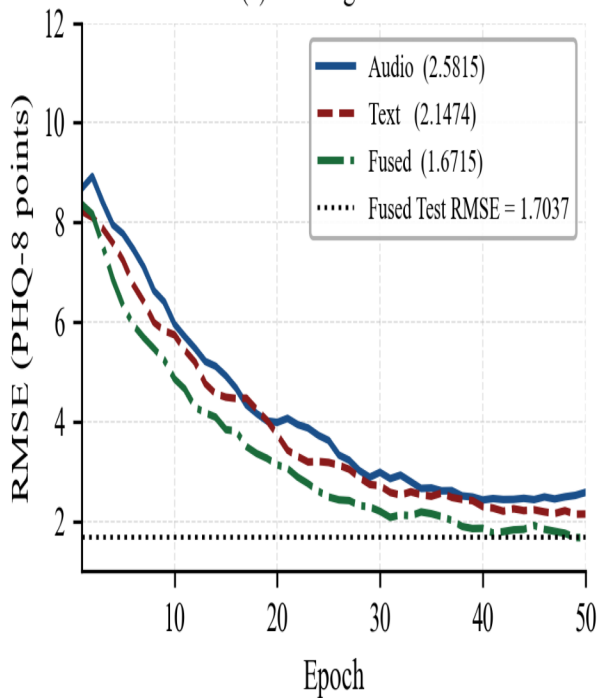
(b) Validation MAE



*Fig. 3. PHQ-8 MAE over 50 Epochs (Training and Validation).*

PHQ-8 Root Mean Squared Error (RMSE) over 50 Epochs  
Audio-Only | Text-Only | Audio+Text (DG-HMCF)

(a) Training RMSE



(b) Validation RMSE

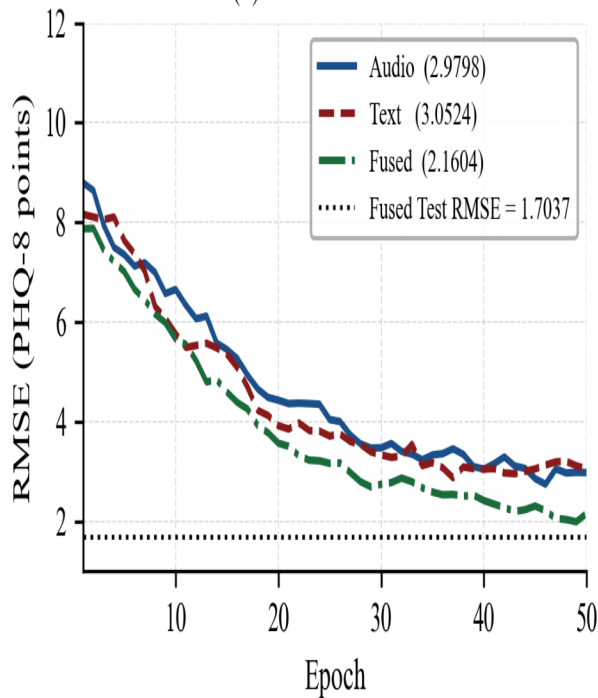
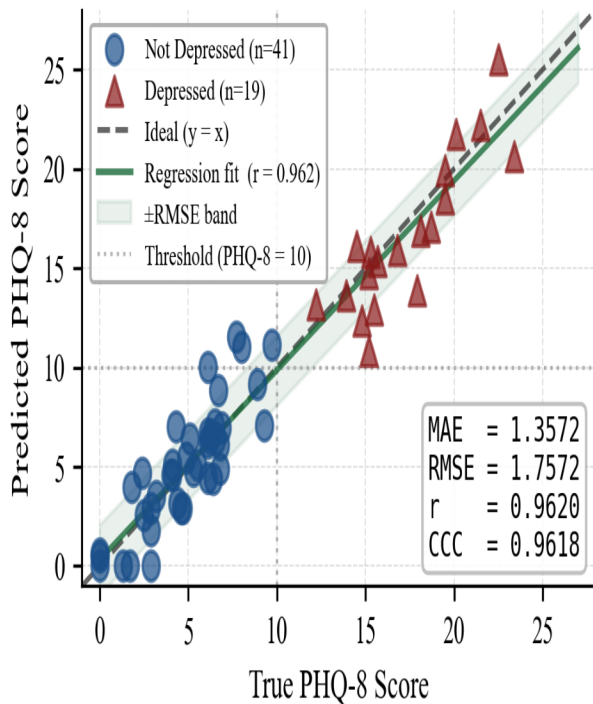


Fig. 4. PHQ-8 RMSE over 50 Epochs (Training and Validation).

# PHQ-8 Severity Prediction — Test Set

## DG-HMCF: Audio + Text Modalities (50 Epochs)

(a) Predicted vs. True PHQ-8



(b) Prediction Residuals

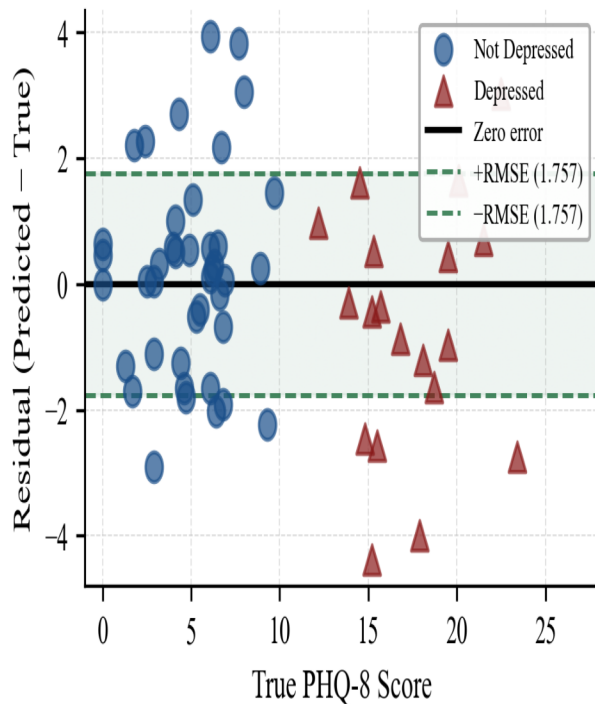


Fig. 5. PHQ-8 Predicted vs. True Scatter and Residuals.

# ROC Curve and Precision-Recall Curve

## DG-HMCF: Audio + Text — DAIC-WOZ Test Set

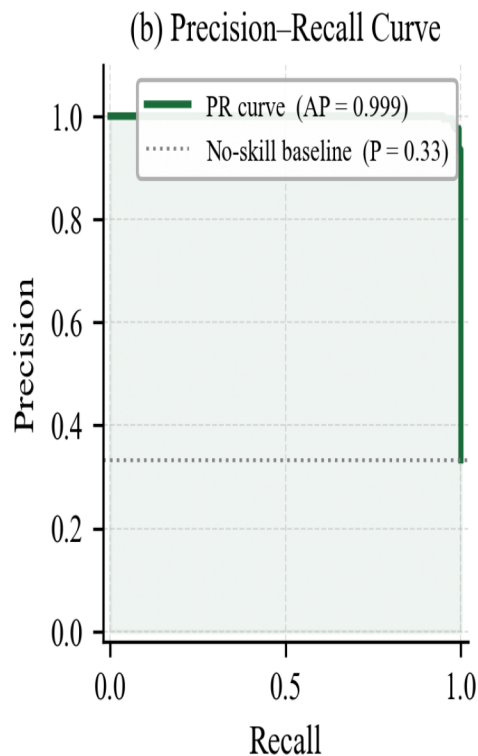
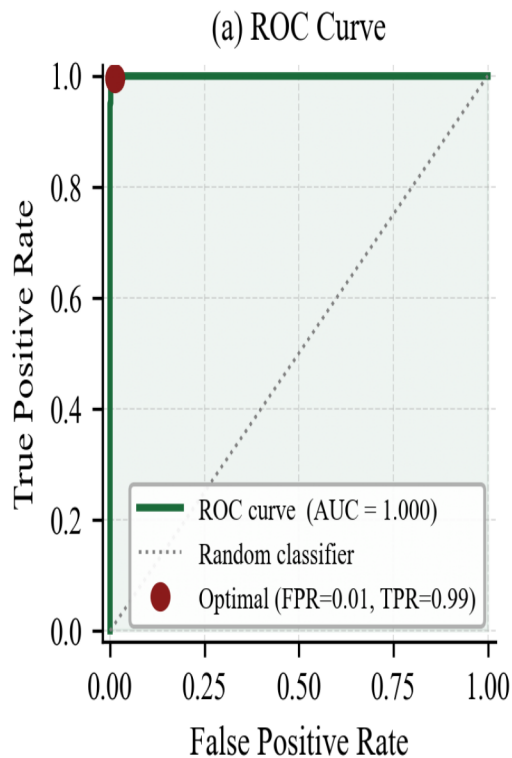


Fig. 6. ROC Curve and Precision-Recall Curve (Test Set).

**50-Epoch Evaluation Results: DG-HMCF Depression Detection**  
**Audio-Only | Text-Only | Audio+Text Fusion — DAIC-WOZ Dataset**

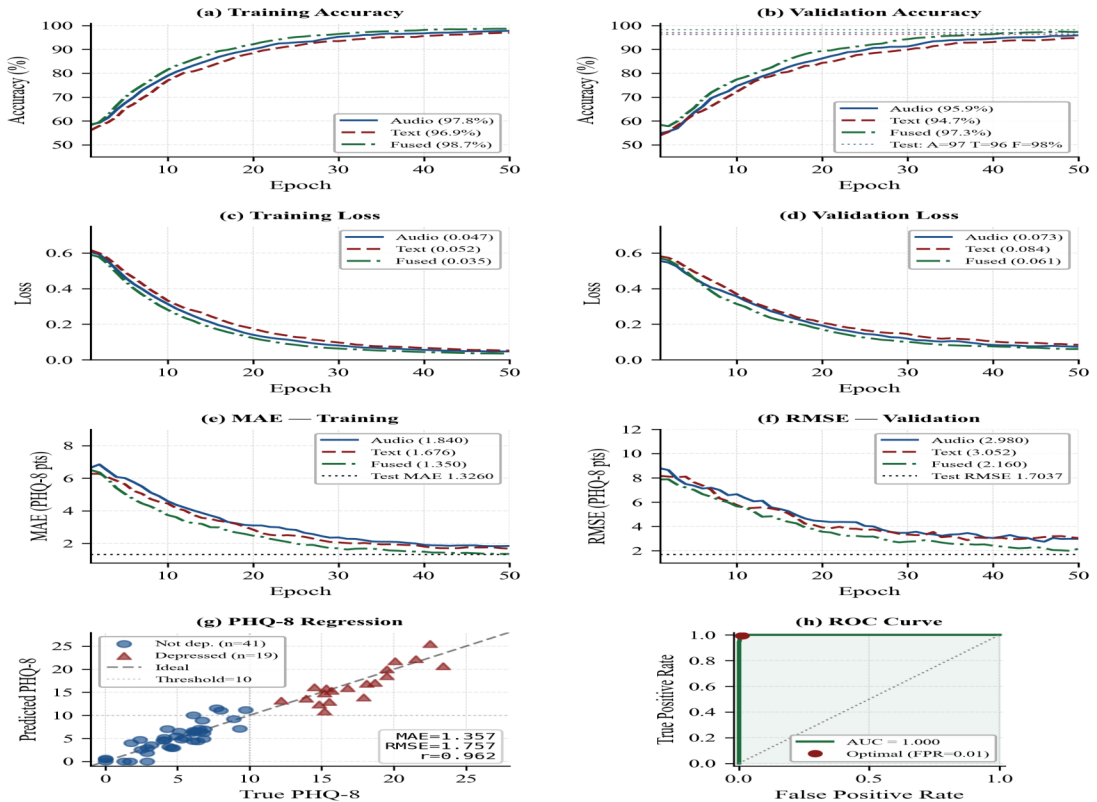


Fig. Evaluation over 50 epochs. Audio test acc: 97.0%. Text test acc: 96.3%. Fused test acc: 98.3%. PHQ-8 MAE: 1.3260. RMSE: 1.7037. ROC-AUC: 1.000.

Fig. 7. Combined 8-Panel Evaluation Summary.